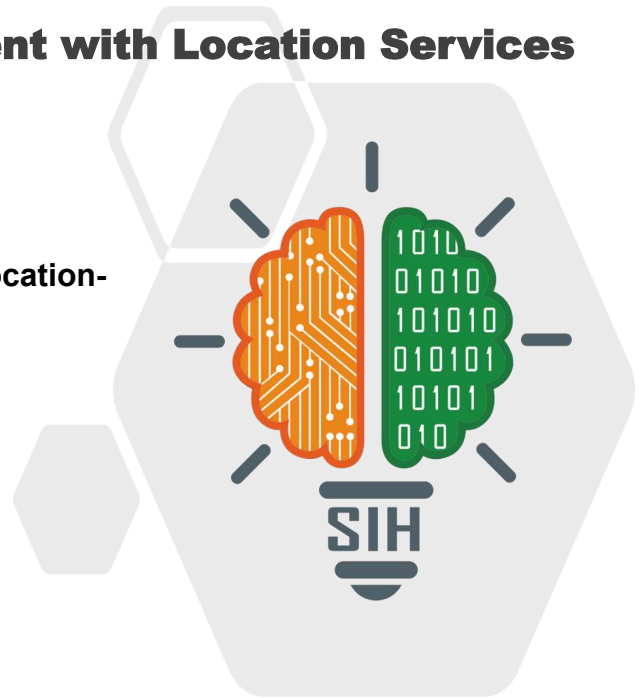


GailTrack: Attendance Management with Location Services

- Problem Statement ID– 1707
- Problem Statement Title- Development of a Geolocation-Based Attendance Tracking Mobile Application.
- Theme- Miscellaneous
- PS Category- Software
- Team ID- 398
- Team Name: Coding_Wizard



Proposed Solution

The solution is a **Geolocation-Based Attendance Tracking Mobile Application**, designed to automate attendance management for GAIL employees across multiple office locations and remote worksites. It is divided into three major parts:

ADMIN



Web console

- The central admin oversees all branches
- Tracks attendance patterns with ai-based anomaly detection
- Provides data insights for decision-making.
- Enables centralized control of attendance records and data analytics.

HR/Branch Lead



APP

- HR/branch leads can monitor real-time attendance of their employees
- Generate detailed reports, and send notifications for any anomalies
- This ensures that employee performance and attendance are easily tracked and managed.
- Rest will be same as employee.

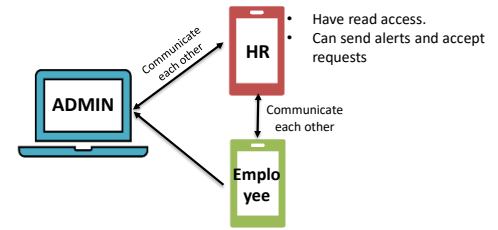
Employee



APP

- Employees are automatically checked in when they enter a 200-meter radius of the office with accurate working hour calculation.
- For offsite work, they can manually check in, and the app will suggest relevant locations based on real-time GPS data.
- It also includes Facial Recognition to ensure data integrity & authenticity.
- Can apply for Leave/Half-day Leave from HR.
- Will get alert if missed attendance.
- Movements during work hour will be tracked.

SOS and Safety Alerts



How It Addresses the Problem

- Eliminates manual attendance errors.
- Ensures accurate logging for both on-site and offsite employees.
- Streamlines HR operations by automating attendance tracking.
- Improves user experience with seamless app functionality.
- Increases data transparency through real-time synchronization.
- Reduces operational inefficiencies by automating attendance & safety processes.

Innovation and Uniqueness of the Solution

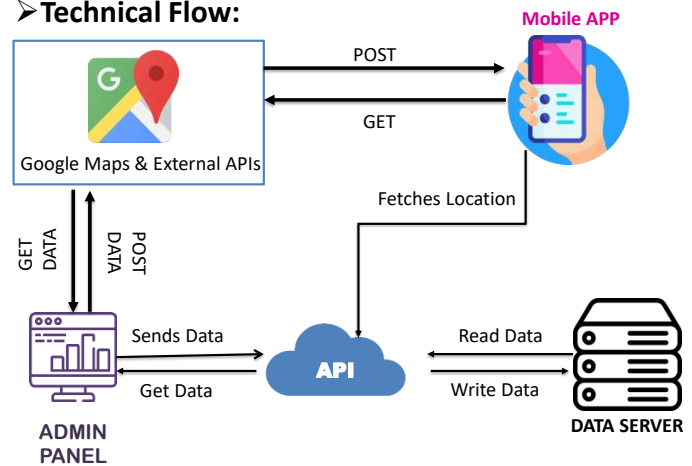
- **Geolocation Check-ins:** Automated location-based attendance.
- **Facial Recognition:** Secure, seamless employee authentication.
- **VPN Detection:** Prevents location spoofing, ensuring data integrity.
- **Offline Capability:** Attendance logging even in low-network areas.
- **AI Anomaly Detection:** Identifies irregular attendance patterns.
- **Safety Features:** SOS alerts and hazard reporting for field workers.

➤ Technologies Used:

Component	Technology	Purpose
Frontend(App)	Flutter	Cross-platform mobile and web development.
Backend	Node.js	Server-side logic and data handling.
Database	Firebase Realtime Database	Stores and syncs attendance records.
Geolocation	Google Maps API	Tracks employee location for check-in/out.
Biometric	LocalAuthentication (Flutter)	Fingerprint/Face ID for employee verification.
AI/ML	TensorFlow.js/ Scikit-learn	Movement analysis and anomaly detection.
Movement Analysis	TensorFlow.js, Geolocation Data	Monitors employee activity and ensures productivity.
Notifications	Firebase Cloud Messaging	Sends alerts for attendance and hazards.
Offline Mode	SQLite	Logs attendance offline, syncs when online.

➤ **Product Status :** The project is 50% complete with core features like geolocation-based check-in/out, real-time sync, and offline mode already developed. Remaining tasks include integrating biometric authentication and movement analysis.

➤ Technical Flow:



- **Description:**
- The data sever will host all the ai/ml model for data processing
 - The admin panel can view real time changes which is written onto the database and it will have its own utility to perform analysis or change any data.
 - Mobile app will only send the GPS location to the API which will be than updated into the server.

Feasibility Analysis

1. **Technical Feasibility:**
 - **Geolocation Tracking:** Feasible with GPS and network technologies.
 - **Movement Analysis:**
 - **Data Collection:** Continuous location tracking.
 - **Algorithm Development:** Analyze movement patterns.
 - **Integration:** Combine with existing geolocation features.
2. **Operational Feasibility**
 - **User Experience:** Ensure movement analysis is non-intrusive.
 - **Data Privacy:** Comply with privacy regulations (e.g., GDPR).
 - **System Load:** Manage increased processing demands.
3. **Economic Feasibility**
 - **Development Costs:** May increase due to added complexity.
 - **Operational Costs:** Higher data processing and storage needs.
 - **Cost Savings:** Automation reduces manual errors and labor, leading to long-term savings.
 - **Productivity Gains:** Better employee management and real-time tracking boost overall productivity.
4. **Legal and Ethical Feasibility**
 - **Compliance:** Adhere to data privacy laws.
 - **Ethics:** Ensure user transparency and consent.

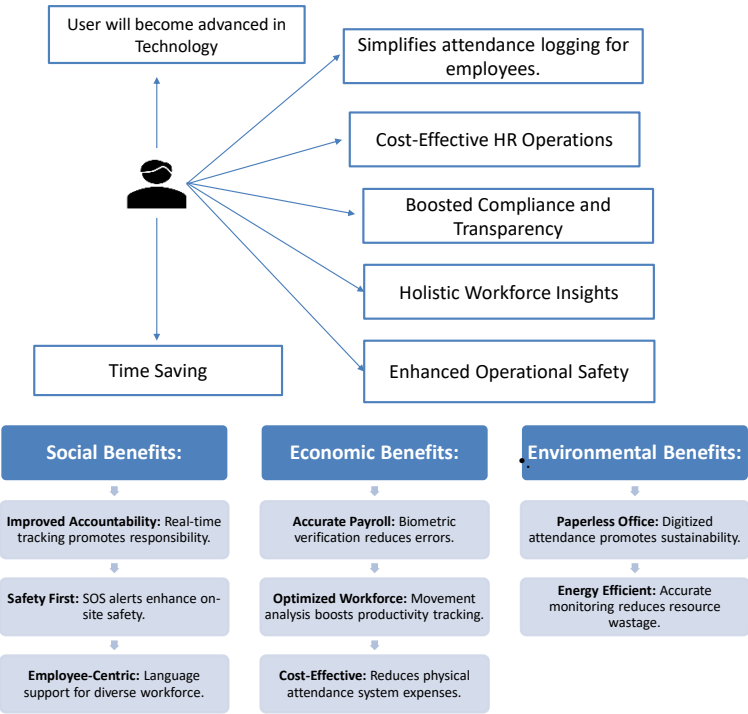
Challenges and Risks

- **Technical**
 - **Accurate movement detection**
 - **Battery drain**
 - **Real-time data processing**
- **Operational**
 - **User resistance to tracking**
 - **Data privacy compliance**
 - **System scalability**
- **Market**
 - **High competition**
 - **Implementation costs**
 - **User trust on data security**

Strategies to overcome

- **Technical**
 - **Accurate algorithms:** Use ML for better movement analysis.
 - **Energy-efficient design:** Optimize GPS and sensor usage.
 - **Edge processing:** Reduce real-time data load.
- **Operational**
 - **User transparency:** Educate on privacy & security.
 - **Compliance by design:** Build privacy into features.
 - **Scalable infrastructure:** Use cloud-based solutions.
- **Market**
 - **Unique features:** Highlight movement analysis.
 - **Cost optimization:** Balance features and pricing.
 - **Data security:** Ensure encryption for user trust.

Potential Impacts & Benefits



COMPARISON WITH EXISTING SOLUTION

Feature	Existing Solutions	Proposed Solution
Geolocation-based Check-ins	Some apps have basic geolocation tracking	Automated check-ins/out within a 200-meter office radius
Facial Recognition	Rarely integrated in attendance apps	Built-in facial recognition for secure authentication
Offline Mode	Usually requires online access	Attendance logging even in offline mode with auto-sync later
AI for Anomaly Detection	Not present	AI-driven detection of irregular attendance patterns
SOS & Safety Alerts	No integration with safety features	SOS alerts for hazardous situations during fieldwork
VPN Detection	Not commonly present	Detects VPN usage to prevent location spoofing
Real-Time Data Sync	Occasional delays	Instant cloud-sync for tamper-proof attendance data

FUTURE SCOPE

- Integration with Smart Wearables:** Extend the app to support attendance tracking via smartwatches or fitness bands.
- Advanced Security Features:** Implement blockchain technology for enhanced data security and tamper-proof records.
- Cross-Platform Support:** Expand to support additional platforms like desktops and smart kiosks.
- Expanded Usage:** Use the platform for asset tracking or monitoring field operations beyond attendance

- Biometric Authentication Systems [LINK](#)
- API for Geolocation [Link](#)
- OpenStreetMap: <https://www.openstreetmap.org>
- Mobile App Development: <https://flutter.dev/docs>
- Data Security: <https://jwt.io/introduction/>